DISTANCE EDUCATION

M.Sc. (Computer Science) DEGREE EXAMINATION, MAY 2012.

MATHEMATICAL FOUNDATION OF COMPUTER SCIENCE

(2007 onwards)

Time : Three hours Maximum : 100 marks

SECTION A — ($10 \times 3 = 30$ marks)

Answer ALL questions.

1. Write the truth table for $p \to q$.

2. Define the term disjunction. Give examples.

3. What is cardinality of a set? Give an example.

4. How can the matrix for a relation be used to determine whether the relation is irreflexible?

5. Define objective. Give an example.

6. Let $f : R \to R$ and $g : R \to R$ where R is the set of real numbers find fog if $f(x) = x^2 - 2$ and $g(x) = x + 4$.

7. Define Semigroup.

8. Define Group.

9. What is complete symmetric digraph? Give an example.

10. Define adjacency matrix. Give an example.

SECTION B — (4 × 10 = 40 marks)

Answer any FOUR questions.

11. Construct the truth table for

$(P \cap R) \cup (\rceil P \cap Q) \cup (P \cap \rceil Q) \cup (\rceil P \cap \rceil Q)$.

12. Show that $R \to S$ can be derived from the premises $P \to (Q \to S)$, $\rceil R\, UP$ and Q.

13. Let R be the relation represented by the matrix $M_R = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$ find the matrices that represent

(a) R^2

(b) R^3

(c) R^4.

14. Prove that the inverse of an a bijections is also a bijection.

15. Prove that for any commutative monoid $(M, *)$ the set of idempotent elements of M forms a sub monoid.

16. Prove that a simple graph with n vertices and 'K' components can have at moot (n – k)(n – k + 1) /2 edges.

SECTION C — (2 × 15 = 30 marks)

Answer any TWO questions.

17. Prove that $(\exists x)(p(x) \cap Q(x)) \Rightarrow (\exists x)\, p(x) \cap (\exists x)\, Q(x)$.

18. Show that $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$, using characteristics function.

19. State and prove the Lagrange's theorem.

DISTANCE EDUCATION

M.Sc. (Computer Science) DEGREE EXAMINATION,
MAY 2012.

COMPUTER ARCHITECTURE

(2007 onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (10 × 3 = 30 marks)

Answer ALL questions.

1. What do you mean by the performance of a computer?

2. Explain Memory Read and Memory Write.

3. Define dynamic scheduling.

4. What is meant by hazards?

5. What are the limitations of ILA?

6. What is meant by static branch prediction?

7. Define miss rate and locality of reference.

8. What is RAID?

9. What do you mean by memory consistency?

10. What is meant by synchronization in multiprocessors?

SECTION B — (4 × 10 = 40 marks)

Answer any FOUR questions.

11. Write short notes on:

    (a) Instruction Format.

    (b) Operand and Operation.

    (c) Control Flow Instructions.

    (d) Encoding an Instruction Set.

12. Discuss on Dynamic H/W prediction and Hardware based speculation.

13. Expound on static multiple issues on ILP.

14. Describe the types of storage devices in detail.

15. Explicate on the models of memory consistency.

16. Explain about RAID.

SECTION C — (2 × 15 = 30 marks)

Answer any TWO questions.

17. Explain the architecture of MIPS and TM32.

18. Discuss in detail on software approaches for Instruction Level Parallelism.

19. Expound on main memory concepts.

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DE–3987 **13**

DISTANCE EDUCATION

M.Sc. (Computer Science) DEGREE EXAMINATION,
MAY 2012.

DATA STRUCTURES USING C++

(2007 onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (10 × 3 = 30 marks)

Answer ALL questions.

1. What is meant by base address of an array? Illustrate.

2. Differentiate "Public" from "Private" related to class declaration in C++.

3. Name three actions that will be performed by pop function in a stack.

4. Write down the infix and postfix notations for summing up two variables A and B.

5. Define the term "Null Pointer".

6. What is a doubly linked list?

7. State the steps of postorder tree traversal.

8. Define : Dangling Pointer.

9. What is meant by order preserving?

10. Define the method, "partition exchange sort".

SECTION B — ($4 \times 10 = 40$ marks)

Answer any FOUR questions.

11. Give a note on character strings in C. Also develop a C program to find the length of the given string.

12. Write a C program to find the sum of above diagonal, below diagonal and trace of a square matrix using two dimensional arrays concept.

13. With syntax, explain the procedure of queue operations and implementation.

14. How does an element is inserted in a doubly linked list? Briefly discuss with algorithm.

15. What are primitive operations on a circular List? Explain.

16. Narrate the efficiency of binary search free operations.

SECTION C — ($2 \times 15 = 30$ marks)

Answer any TWO questions.

17. Write a C program to convert an expression from infix to postfix.

18. Explain the concept of Queue and its sequential representation with illustrations.

19. With program segment, explain the procedure of Merge sort.

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DE–3988 **14**

DISTANCE EDUCATION

M.Sc. (Computer Science) DEGREE EXAMINATION,
MAY 2012.

VISUAL PROGRAMMING

(2007 onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (10 × 3 = 30 marks)

Answer ALL questions.

1. What is an event in windows programming?

2. What is meant by resource?

3. Specify the difference between command button and label in VB.

4. List out various items in the toolbar.

5. Mention any three properties of Grid control.

6. What is the use of data control in VB?

7. What do you mean by bit map?

8. What is an object?

9. Expand ODBC and DLL.

10. Give any three DAO database classes in VC++.

SECTION B — (4 × 10 = 40 marks)

Answer any FOUR questions.

11. Give an overview of windows programming environment and highlighting its salient features.

12. Explain (IF...ELSE, SELECT...CASE, SWITCH, FOR, WHILE...LOOP) with examples.

13. State the properties of Richtext box control and compare with textbox with examples.

14. (a) Describe the steps in creating a menu in VC++. (5)

    (b) Explain the steps in creating a Dialog box in VC++. (5)

15. Write down the steps and explain each of them to add ODBC database capability to an MFC application.

16. What are the constituents of a dynamic Link Library? How it is different from a runtime library? What are its strengths?

SECTION C — (2 × 15 = 30 marks)

Answer any TWO questions.

17. Write in detail about procedures and functions in VB.

18. Design and implement student database in VB.

19. (a) Explain in detail about serialization in VC++. (7)

    (b) Explain the use and purpose of MDI in VC++. (8)

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DE–3989 **15**

DISTANCE EDUCATION

M.Sc. (Computer Science) DEGREE EXAMINATION,
MAY 2012.

DATABASE MANAGEMENT SYSTEMS

(2007 onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (10 × 3 = 30 marks)

Answer ALL questions.

1. How does base differs from a data file?

2. Give the purpose of secondary key.

3. Mention the strengths of relational model.

4. How the mapping operation is done on the relational model?

5. Define Client/Server system.

6. How physical file differs from logical file?

7. Define objects.

8. Give an example for Oracle Query.

9. What is meant by Index?

10. What is meant by Embedded SQL?

SECTION B — (4 × 10 = 40 marks)

Answer any FOUR questions.

11. Discuss about the concept of database schema.

12. Mention the salient features of database management system.

13. What are the semantic issues that arise in relational model?

14. List the benefits of Oracle.

15. How to select distinct values through SQL?

16. Explain the techniques of modifying tables.

SECTION C — (2 × 15 = 30 marks)

Answer any TWO questions.

17. Discuss about the different user interfaces in DBMS.

18. Explain the following in relational model

    (a) Data definition facilities

    (b) Data control facilities.

19. Discuss about the languages supported by oracle pre compiler in Embedded SQL.

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DE–3990 **21**

DISTANCE EDUCATION

M.Sc. (Computer Science) DEGREE EXAMINATION,
MAY 2012.

COMPUTER NETWORKS

(2007 onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (10 × 3 = 30 marks)

Answer ALL questions.

1. Discuss about N/W Hardware.

2. Write about LAN.

3. Write a note on error correcting codes.

4. Discuss about PPP.

5. Explain Tunneling.

6. What do you mean by congestion?

7. Explain about TCP and layers of TCP/IP.

8. Write short notes on crash recovery.

9. Define cryptography.

10. What do you mean by multimedia?

SECTION B — (4 × 10 = 40 marks)

Answer any FOUR questions.

11. Explain OSI Reference model.

12. Discuss on communication satellites.

13. Discuss about HDLC and SLIP protocols.

14. Describe about the ALOHA and collision-Free protocols.

15. Explain about UDP.

16. Write a detailed note on Secret and public key algorithms.

SECTION C — (2 × 15 = 30 marks)

Answer any TWO questions.

17. Discuss on
    (a) ATM.
    (b) Communication Satellites.

18. Discuss an a Routing algorithms.

19. Explain about
    (a) Data compression
    (b) Client side and server side.

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DE–3991 **22**

DISTANCE EDUCATION

M.Sc. (CS) DEGREE EXAMINATION, MAY 2012.

SOFTWARE ENGINEERING

(2007 onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (10 × 3 = 30 marks)

Answer ALL questions.

1. What are the steps involved in the water fall model and why it is called so?

2. What do you mean by software process?

3. Write a note on project estimation.

4. What do you mean by COCOMO model?

5. What are the projects that deals with the software organization encounter?

6. What do you mean by quality standards?

7. What is system engineering and give the hierarchy of system engineering?

8. Expound on black box testing.

9. Discuss on Data design.

10. Define Debugging.

SECTION B — (4 × 10 = 40 marks)

Answer any FOUR questions.

11. Write a detailed notes on spiral model.

12. Explain the methodology to conduct the formal technical review.

13. Discuss about risk management.

14. Write short notes on s/w quality assurance.

15. Explicate on design concepts and principles.

16. Expound on testing strategies.

SECTION C — (2 × 15 = 30 marks)

Answer any TWO questions.

17. Discuss in detail about the evolutionary s/w process model and 4G techniques.

18. Write a detailed notes on

    (a) S/w reviews.

    (b) Specification modeling and information flow.

19. Discuss in detail about s/w project planning.

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DE–3992 | **23**

DISTANCE EDUCATION

M.Sc. (Computer Science) DEGREE EXAMINATION,
MAY 2012.

INTERNET PROGRAMMING AND WEB DESIGN

(2007 onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (10 × 3 = 30 marks)

Answer ALL questions.

1. List out the functions of IP.
2. What is proxy gateway?
3. Compare Java with Javascript.
4. How can you declare variables in VBScript?
5. What are the packages that can be shocked?
6. How ActiveX controls can be added to web pages?
7. What is the use of META element?
8. State the purpose of HIDDEN attribute.
9. How can you insert an image into a web page?
10. Give the advantages of GIF animation.

SECTION B — (4 × 10 = 40 marks)

Answer any FOUR questions.

11. Explain the TCP/IP protocol model.
12. Explain method overloading in Java with an example.

13. Describe the built-in objects and functions available in Javascript.

14. Describe the components of SGML system.

15. How can you add functionality to the skeleton?

16. Describe the steps to add the navigation bars to the web page using dreamweaver.

SECTION C — (2 × 15 = 30 marks)

Answer any TWO questions.

17. Describe the looping statements available in VBScript.

18. Develop a complete web page using frames and framesets which gives the information about a hospital.

19. With suitable example, explain a multi-protocol application on the web.

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DISTANCE EDUCATION

M.Sc. (Computer Science) DEGREE EXAMINATION,
MAY 2012.

IMAGE PROCESSING AND ANALYSIS

(2007 onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (10 × 3 = 30 marks)

Answer ALL questions.

1. Write a note on : Scenes and Images.

2. What are different distance measures?

3. What are Median filters?

4. Write a note on : Image sharpening.

5. What are the major steps of Data compression procedure?

6. What is meant by pseudo-color processing?

7. What are Prewitt and Sobel operators?

8. Write a note on : Multi-level thresholding.

9. What are Topological attributes?

10. Write a note on : Graph isomorphism.

SECTION B — (4 × 10 = 40 marks)

Answer any FOUR questions.

11. Write short notes on : Perception of color.

12. Explain about basic transformations in Geometric model.

13. Discuss about restoration by homomorphic filtering.

14. Explain the Huffman coding process with an example.

15. Write about Morphological edge detection.

16. Explain the deterministic methods in recognition.

SECTION C — ($2 \times 15 = 30$ marks)

Answer any TWO questions.

17. Write in detail about the following :

(a) Brightness adaptation and contrast. (8)

(b) Visual details in the digital image. (7)

18. Explain about Multi-valued image processing.

19. Discuss the following :

(a) Optimum threshold. (8)

(b) Fuzzy mathematical recognition. (7)

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DE–3994 **25**

DISTANCE EDUCATION

M.Sc. (Computer Science) DEGREE EXAMINATION,
MAY 2012.

OPERATING SYSTEMS

(2007 onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (10 × 3 = 30 marks)

Answer ALL questions.

1. Give the purpose of operating system.

2. Define system call.

3. What is meant by deadlock?

4. Mention the concept of Thread.

5. What is meant by Swapping?

6. How paging is done?

7. Name any three access methods for file systems.

8. How to protect the files?

9. Write the concept of Event Ordering.

10. List any three salient features of LINUX.

SECTION B — (4 × 10 = 40 marks)

Answer any FOUR questions.

11. Explain the concept of clustered systems.

12. How CPU scheduling is done?

13. Discuss on the Inter process communication in process management.

14. Mention the strategies involved in Continuous Memory allocation.

15. Discuss on the directory implementation of the file system.

16. How to obtain the concurrency control in distributed operating system?

SECTION C — (2 × 15 = 30 marks)

Answer any TWO questions.

17. Explain the following systems:

    (a) Multiprocessor system

    (b) Hand held systems

    (c) Main frame system.

18. Discuss on the methods for handling dead locks.

19. Compare and contrast stateful services with stateless services in distributed Operating System.

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DE–4398 **16**

DISTANCE EDUCATION

M.Sc. (Computer Science) DEGREE EXAMINATION, MAY 2012.

LAB I — C++ AND DATA STRUCTURES

(2007 onwards)

Time : Three hours Maximum : 100 marks

Break-up of marks :

Record : 10
Algorithm : 10
Program : 50
Debugging : 10
Execution : 10
Result : 10
Total : 100

Examiner has to select and give ONE question to each candidate by lot system.

1. (a) Develop a C++ program to find the factorial of a given number using recursive function.

   (b) Write a C++ program to sort the given set of elements using quick sort.

---------- Cut here ----------

2. (a) Write a C++ program using polymorphism concept to calculate the square of any two numbers of type int, float, double and long.

   (b) Develop a C++ program to sort the given set of elements using insertion sort.

3. (a) Develop a menu driven C++ program to add and subtract the given two matrices of order $m \times n$ defined in a class, using operator overloading.

(b) Write a C++ program to sort the given set of elements using merge sort.

---------- Cut here ----------

4. (a) Write a C++ program to find all possible roots of a given quadratic equation using function.

(b) Write a C++ program to implement insert and delete operations in doubly linked list.

---------- Cut here ----------

5. (a) Write a C++ program to read 10 values, to an array variable. Use pointers to locate and display each value.

(b) Develop a C++ program to implement push and pop operations on a stack.

---------- Cut here ----------

6. (a) Develop a C++ program to print a triangle of star symbols.

```
      *
    * * *
  * * * * *
* * * * * * *
```

(b) Write a C++ program for linked list implementation of queue operations.

7. (a) Develop a C++ program to find whether the given string is palindrome or not.

(b) Write a C++ program to sort the given set of elements using Bubble sort.

---------- Cut here ----------

8. (a) Develop a C++ program with operator overloading concept to find the difference and total length of given two various tubes specified in metres and centimetres.

(b) Write a C++ program to implement insert and delete operations in doubly linked list.

---------- Cut here ----------

9. (a) Develop a C++ program to sort 10 numbers in ascending order with naming of variable and the value before and after sorting.

(b) Write a C++ program to evaluate the given mathematical expression using stack.

---------- Cut here ----------

10. (a) Write a C++ program for matrix operations using functions.

(b) Develop a C++ program to sort the given set of elements using merge sort.

DE–4399 **17**

DISTANCE EDUCATION

M.Sc. (Computer Science) DEGREE EXAMINATION,
MAY 2012.

COMPUTER LAB — II (VISUAL PROGRAMMING)

(2007 onwards)

Time : 3 hours

Maximum : 100 marks
Record: 10
Algorithm: 10
Program : 50
Debugging : 10
Execution : 10
Result : 10
Total : 100

Candidate has to select ONE question by lot system.

Answer BOTH Subdivisions (a) and (b)

1. (a) Write event procedure to perform following on string.

   (i) Reverse

   (ii) Palindrome or not

   (iii) Change case.

   (b) Write VC++ program to fill background of the client area with a bit map.

2. (a) Using Active X control create a Text box that accepts only numeric value with the given properties.

   (i) Background of Text box.

   (ii) Foreground of Text box.

   (iii) Text property of Text box.

   (iv) Resize the text box without using properties.

   (b) Write a VC++ Program to get the status of shift and toggle keys using MFC.

---------- Cut here ----------

3. (a) Write a VB program to perform the following operation in a record of random Allen File.

   (i) Insert.

   (ii) Delete.

   (iii) Search.

   (b) Write a VC++ program to create a List box in a window.

---------- Cut here ----------

4. (a) Using MsFlex grid display the multiplication and addition table of 20 rows and columns.

   (b) Write a VC++ program to find out whether a mouse is Attached or not, and if attached how many buttons are present or not.

---------- Cut here ----------

5. (a) Using Control array create a simple arithmetic calculator to do the operations like addition, subtraction, Multiplication, division, Square, Square root, mod, power, Percentage.

   (b) Write a VC++ program to create a List box in a Window.

6. (a) Using built in AX control, develop the windows Notepad with File ad Edit menu and display floating men whenever necessary.

(b) Write a VC++ program to get the status at the shift and toggle keys using MFC.

---------- Cut here ----------

7. (a) Create a table Hospital with patient id, name, addr, B Group, Disease and use Ax DAO to develop Hospital management system to insert, search, delete, modify and display the records with corresponding B Group form combo box.

(b) Write a VC++ program to generate a status bar and show the status of caps Lock, Num Lock and Scroll Lock in it.

---------- Cut here ----------

8. (a) Create a VB application with a Drive List box Dir List Box.

(i) Select exe file and execute it.

(ii) Select the picture and load it to the form.

(iii) Filter the file in the File list box according to the extension chosen in the combo box.

(b) Write a VC++ program to fill back ground of the client area with a bitmap.

9. (a) Develop a data report using employee table with the following:

(i) Display employee details.

(ii) Dept. wise emp details and total salary in each department.

(iii) Display employee details which starts with name specified in Text box.

(b) Write a VC++ program to handle windows messages in MFC program.

---------- Cut here ----------

10. (a) Create a form with Text box, combo box, and command Button and do the following:

(i) Add university name in the combo box at run time.

(ii) Search and delete the particular university in the combo box.

(iii) Display the message box with number of university available in the combo box.

(iv) Sort names of universities in Ascending order.

(b) Write a VC++ program to create a window of desired size using MFC.

DE–4400 **26**

DISTANCE EDUCATION

M.Sc. (Computer Science) DEGREE EXAMINATION,
MAY 2012.

COMPUTER LAB — III
(INTERNET PROGRAMMING LAB)

(2007 onwards)

Time : Three hours Maximum : 100 marks

Examiner has to choose any ONE question.

Break-up marks

Record : 10
Algorithm : 10
Program : 50
Debugging : 10
Execution : 10
Result : 10
Total : 100

---------- Cut here ----------

1. (a) Design and publish a web page for your college.

   (b) Write a Java script to create a window by using the confirm message.

---------- Cut here ----------

2. (a) Develop a web page for job recruitment agency in an IT Industry.

   (b) Write a VB script to do the following :

      (i) Check the given password.

      (ii) Change the existing password.

3. (a) Develop a single page advertisement for a shop to be opened shortly.

(b) Write a menu drive Java program to do database functions using database connectivity facility. The functions include display, edit, insert and delete a record from a table.

---------- Cut here ----------

4. (a) Write a Java program using Applet to display any three images when three buttons in the border layout are clicked. The image should be displayed at the centre.

(b) Write a VB script to create a calender for a given month and year.

---------- Cut here ----------

5. (a) Using atleast 20 HTML Tags, create a screen with a string "WEB DESIGN".

(b) Write a Swing program to create buttons with

(i) Tool top text

(ii) Image

(iii) Border

(iv) Shortcut key.

---------- Cut here ----------

6. (a) Write a Java Program using Applet to create the frames and its controls.

(b) Write a VB Script to do the following :

(i) Display the current date.

(ii) Find the difference between the two dates.

(iii) Find the age of a person by providing date of birth.

7. (a) Develop a picture gallery having atleast 5 pages. Each of them is having several pictures with suitable information.

(b) Write a Java Script to create a order form to select the house articles.

---------- Cut here ----------

8. (a) Write a Java program using Applet to display the different colors and fonts.

(b) Write a S-wing program to create the Tabbed Panels.